MOHSIN IBNA HOSSAIN
AIUB, IP NOTES MID TERM

# Introduction To C++

TOPIC NAME :

DAY :

TIME : DATE : / /

# Table of Contents


| SL.NO | Topic | Pages |
|---|---|---|
| 1 | Data Types | 03 |
| 2 | Introduction to C++ | 07 |
| 3 | Type Casting | 20 |
| 4 | Conditional Statement | 24 |
| 5 | Multiple Condition | 33 |
| 6 | Nested if/else | 35 |
| 7 | Logical Operators | 37 |
| 8 | Switch Case | 40 |
| 9 | for loop | 44 |
| 10 | while loop | 50 |
| 11 | do while loop | 52 |
| 12 | Swap operation | 56 |
| 13 | Continue & break | 57 |
| 14 | Nested for loop | 58 |
| 15 | array | 65 |

# Calculate the average of number 2:—

input a, b

Average = (a+b)/2 → Constants

Ex:- 3, 3.14, 'e'

(a+b) → Variables

Memory:—

$\boxed{20}$ a

4byte

$\boxed{30}$ b

4byte

=> কোনো Variable memory তে কতটুকু জায়গা নেবে তা Data Type এর উপর নির্ভর করে।

## #Data Types:—

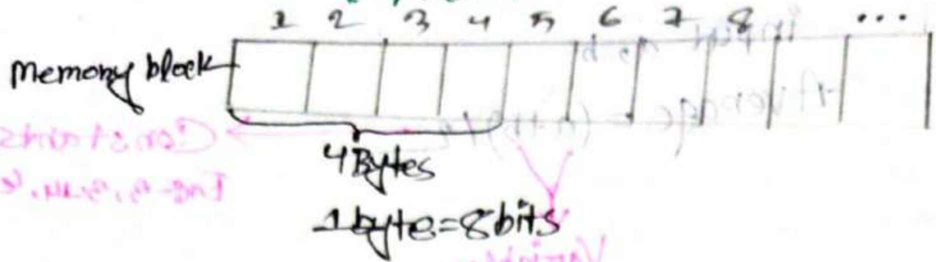
=> Data types ৩ ধরনের হয়।

1. Primitive Data type = build in Data type.
2. Derived Data type = Kind of build in Data type.
3. User-defined = আমরা As a user define করি।

## # Primitive Data Types:—

=>
1. Integer → 1, 4, 100, -100.
2. Float → 3.14, 6.5, 1.00
3. Character → c, f, @, #, %
4. Boolean → 0, 1

## Integer:—

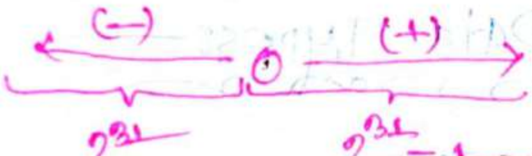



1 byte = 8 bits

তাহলে যেকোনো integer 4*8 = 32 Bits জায়গা দখল করে memory তে

for positive int:—
এক্ষেত্রে Range (unsinged) = 0 to $2^{32} - 1$

for nagative int:—
এক্ষেত্রে Range ( singed) = $2^{31}$ to $2^{31} - 1$




## FLOAT:—

Size = 4 byte

Ex:- 4.322, 5.16 (up to 7 decimal digits)

যদি 7 decimal digits থেকে বেশি digit ব্যবহার করতে চাই তাহলে সেক্ষেত্রে আমরা Double ব্যবহার করবো।

## * DOUBLE:—

Size: 8 byte (up to 15 decimal digits)

## CHAR:—

Size: 1 byte.

int ascii = 'a';

ascii -> 97

Rules:— Chanten লেখার সময় (' ') single কোটেশন ব্যবহার করতে হবে।

## BOOL:—

Stores boolean value -> true and false.

Size = 1 byte.

# Now See How to write Variable:—

```
# include<iostream>
using namespace std;
int main() {
        int a; //Variable declaration.
        a=12; //initialisation.

        Cout<<"size of int"<<sizeof(a)<<endl;

        float b;
        b = 13.5;
        Cout<<"size of flo at"<<Size of(b)<<endl;
```

```
char c;
cout<<"size of char"<<sizeof(c) <<endl;

bool d;
cout<<"size of bool "<<sizeof(d) <<endl;

return 0; }
```

**Output:-**

Size of int 4
Size of float 4
Size of char 1
Size of bool 1

note:- Sizeof এটি একটি build function. যা দিয়ে যেকোনো Variable এর size Auto বের করতে পারে।

# Type Modifiers:-

Signed → 4 bytes → -2,147,483 2,147,647
Unsigned → 4 bytes → 0 to 4,294,967,295
long → 8 bytes → -9,223,372 to +9,223,372
Short → 2 bytes → +32,768 to 32,767.

```
# include <iostream>
using name space std;
int main() {
    Short int si;
    long int li;
    Cout << "size of short int" << sizeof(si) << endl;
    Cout << "size of long int" << sizeof(li) << endl;
return 0; }
```

Output:-

Size of short int : 2
Size of long int : 8

# C++ body সিনট্যাক্স:

# include <iostream> → Header file for taking input and printing output

Preprocessor ~~Dec~~ Directive
Used to include files

Using namespace std;

int main() {

main → The execution of code begins from main function.

Used to display output in quotation marks ← Cout << "Hell World" << endl; → marks the end of a statement

endl → adds line break

return 0;

return → Exit Status of a function

} → Curly braces indicate the start and end of a function.

১. Std = অনেক গুলো word এক সাথে Point করার সময় দেওয়া হয়।

২. int = integer (৫, 0, ১০, -২০০, -৫)

৩. return 0; এর মানে হচ্ছে এই function দিয়ে কাজ এখানেই শেষ।

৪. #include <iostream> = Heading tag.

৫. int = Natural numbers বোঝানোর জন্য

৬. endl = লাইনের মধ্য Break Point দেওয়ার জন্য।

৭. main() = এটি একটি main function যেখান থেকে Code Start করা হয়।

৮. // = এটি দিয়ে লাইন Comment করা হয়

৯. */ = এটি দিয়ে অনেক গুলো লেখাকে এক সাথে Comment করা হয়।

# Print any Value:—

```
#include <iostream>
using namespace std;
int main() {
    int a=5;
    float b=5.5;
    cout<<"Value of a: "<<a<<endl;
    cout<<"Value of b: "<<b<<endl;
}
```

Output:

Value of a: 5
Value of b: 5.5

## # DO Sum & Sub:

```
# include<iostream>
using namespace std;
int main(){
    int a=5;
    float b= 5.5;
    float sum =a+b;
    float sub = b-a;
    cout<<"Value of sum:"<< sum<<endl;
    Cout<<"Value of Sub:"<< Sub <<endl;
  return 0;
}
```

Output:

Value of sum: 10.5
Value of sub: 0.5

## # DO mult & div & rem:

```
# include<iostream>
using namespace std;
int main(){
    int a=5;
    float b=2;
    int mult = a*b;
    int div = a/b;
    int rem = a%b;
```

```
cout << "Value of multi:" << mult << endl;
cout << "Value of div:" << div << endl;
cout << "Value of rem:" << rem << endl;
  return 0;
}
```

Output:

Value of mult: 10
Value of div: 2
Value of rem: 1

# Condition:

1. float ÷ int = float
2. int ÷ int = int
3. int ÷ float = float
4. float ÷ float = float

# Ex: 1

```
#include <iostream>
using namespace std;
int main(){
    int a=5;
    float b=a;
  cout<<"b:"<<b<<endl;
  return 0;
}
```

Output:

b: 5

Ex: 02

```
#include <iostream>
using namespace std;
int main(){
    float a= 5.75;
    int b = a
    Cout<<" Value of b:"<<b <<endl;
    return 0;
}
```

Output:

Value of b: 5

# Check the size of float, int, Char, double, bool:

```
#include <iostream>
using namespace std;
int main(){
    int a= 1;
    float b= 2.55;
    duble c= 4.56;
    Char d= 'z';
    bool e= true;
    boll f= false;
    Cout <<"size of a: "<<sizeof(a) <<endl;
    Cout<<"size of b: "<<size of(b) <<endl;
    Cout<<"size of c: "<< size of(c) <<endl;
```

```
Cout<<"size of d:"<<size of (d) <<endl;
Cout<<"size of e:"<<size of (e) <<endl;
Cout<<"size of f:"<<size of (f) <<endl;

return 0;

}
```

Output:

```
Size of a: 4
Size of b: 4
Size of c: 8
Size of d: 1
Size of e: 1
Size of f: 1
```

Disclaimer:

1. কোনো key word কখনোই কোনো Variable এর নাম হতে পারে না।

2. int a } those are not same.
   int A }

3. float } those are not same.
   Float }

4. কোনোই Variable এর নামের শুরুতে Digit ব্যবহার করা যাবে না,

5. কোনোই Variable এর নামের মাঝে Space ব্যবহার করা যাবে না,

6. @, #, -, ,, $ those are spacial char.

7. Variable এর নামের মাঝে কোনোই স্পেশাল char ব্যবহার করা যাবে না except : - (under score)

int _Var_1_1;

# Input/Output :-

Q:- কিং, মেহের মা এবং বাবা তাকে কিছু টাকা দিলো, তাহলে আমাদের বের করতে হবে final ammount টা কত হবে,

```
#include <iostream>
using namespace std;
int main(){
    int amount1;
    int amount2;
    cout<<"Enter your 1st amount:";
    cin>> amount1;
    cout<<"Enter your 2nd amount:";
    cin >>amount2;
    int sum=amount1+amount2;
    cout<<"sum:"<< sum<<endl;
    return 0;
```

Output:—

Enter your 1st amount: 100

Enter your 2nd amount: 150

Sum: 250

# Post increment/Decrement
# Pre increment/Decrement

Condition:—

1. a++ → post increment = a++ = a+1

2. a-- → post Decrement = a-- = a-1

3. ++a → Pre increment = a+1 = ++a

4. --a → Pre Decrement = a-1 = --a

## #Ex: 1

```
#include <iostream>
using namespace std;
int main() {
    int a=5;
    float b=6.58;
    b++;
    int c=b;
    --c;
    a--;
    c=a;
```

```
Cout << "Value of a: " << a << endl;
Cout << "Value of b: " << b << endl;
Cout << "Value of c: " << c << endl;

  return 0;
}
```

a = 5
b = 6.58
b = 6.58 | b = 7.58
c = 7
c = 6
a = 5 | a = 4
c = 4

**Output:**

Value of a: 4
Value of b: 7.58
Value of c: 4

## # Precedence:

### Precedence of Operators:

1. !, ++, --, (type)
2. *, /, %, x
3. +, -
4. =

* একটি Problem এ একই level এর Precedence operators থাকলে সে ক্ষেত্রে left থেকে Right এ হিসেবে হিসাব করতে হবে।

**Ex: 1**

```
# include <iostream>
using namespace std;
int main() {
```

**Output**

Exp: 6

```
int a=5;
int b=2;
int c=3;
int exp1= c+a/b;
int exp2= a/b*c;
Cout<<"Result1:"<<exp1<<endl;
Cout<<"Result2:"<<exp2<<endl;
return 0;
}
```

**Output:**

Result1: 5

Result2: 7

বাবে Precedence:

$exp_1 = a/b = \frac{5}{2} = 2.5+c = 5.5$

$exp_2 = a/b = \frac{5}{2} = 2.5 \times 3 = 7.5$

তখন প্রয়োজন Associativity হয়।

Ex: 2

```
#include <iostream>
Using namespace std;
int main(){
 int a=9;
 int b=5;
 int c=2;
int exp = a-b+c;
Cout<<"Exp: "<<exp<<endl;
}
```

**Output**

Exp: 6

বাবে

$= 9-5 = 4+2 = 6$

Ex 3

```
# include <iostream>
using namespace std;
int main() {
    int a = 4;
    int b = --a;
    cout << "Value of a: " << a << endl;
    cout << "Value of b: " << b << endl;
}
```

Output:-

Value of a = 3
Value of b = 3

a = 4
a = 3 || b = 3

Basic ① a=3;
b=a++;
b=a;
b= 3
a=4

(ii) a=3;
b=a-- ; b=a;
b=3
a=2

a=3;
(iii) b= ++a;
b=a;
b=4
a=4

(iv) a=3;
b= - -a;
b=a;
b= 2
a=2

Ex:- 4

```
# include <iostream>
Using namespace std;
int main() {
  int a=4;
  float b= 7.88 ;
  int c= b--;
  b--;
  a= b;
  c= --b;
  Cout << "Value of a:" << a << endl;
  Cout << "value of b:" << b << endl;
  Cout << "Value of c:" << c << endl;
}
```

Output:-

Value of a: 5

Value of b: 4.88

Value of c: 4

तरीका

a=4

b= 7.88

c=7 // b=6.88

b=5.88

a= 5

b= 4.88 // c= 4

## Ans-5

```
#include <iostream>
using namespace std;
int main() {
  int a = 9;
  float b = 9.48;
  float c = b--;
  ++b;
  a = b++;
  c = --b;
  b = ++c;
  cout<<"a: "<<a <<endl;
  cout<<"b: "<<b <<endl;
  cout <<"c: "<<c<<endl;
}
```

### Output:

a: 9
b: 10.48
c: 10.48

a = 9
b = 9.48
c = 9.48 // b = 8.48
b = 9.48
a = 9 // b = 10.48
b = 9.48 // c = 9.48
c = 10.48 // b = 10.48

Ex:6

```
int main(){
  float var1 = 45.76;
  float var2 = (int) var1;
 cout<<"var1: "<<var1<<endl;
 cout<<"var2: "<<var2<<endl;
}
```

Output:-

Var1 = 45.76
Var2 = 45

# Type Casting:-

⇒ Casting is a mechanism that allows me to convert a variable from one datatype to another.

Ex:-1

```
# include <iostream>
 using namespace std;
 int main(){
  char var1 = 'c';
  char var2 = 'A';
```

```
int sum = (int)var1 + int(var2) - 35;
cout<<"sum:"<<sum<<endl;
}
```

Output:

Sum: 132

Ex:2

```
int main(){
    char var1= 'C';
    var1++;
    cout<<"character: "<<var1<<endl;
}
```

Output:-

Character: D

Ex: 3

```
int main(){
    float a= 5;
    float b= 2;
    float div = a/b;
    cout<<"Div: "<<(int)div<<endl;
}
```

Output:-

Div = 2

Ex: (4) int main(){

```
int a=5;
int b=2;
float div=(float)(a/b);
Cout<<"Div: "<<div<<endl;
}
```

Output:-

Div: 2

Because:-

int ÷ int = int

Ex: 5

```
int main{
float a=2.5;
float b=6.5;
float c=4.5;
float exp= a+ int(b)+(int)c;
Cout<<"Result: "<< exp<<endl;
}
```

Output:-

Result:- 12.5

Ex: (5)

```
int main{
int a=5;
Cout<< "a:"<<++a<<endl;
Cout<<"a :"<<a++<<endl;
Cout<<"a:"<<a<<endl;
}
```

Output:-

a: 6
a: 6
a: 7

Dry run
a=5
a=6
a=6 // a=7
a=7

Ex: (6)

```
#include<iostream>
using namespace std;
int main(){
float a=35.76;
int b=a++;
float c=a;
b=--c;
a++;
Cout<<"a:"<<a<<endl;
Cout<<"b:"<<b<<endl;
Cout<<"c:"<<c<<endl;
}
```

Dry run
a=35.76

b=35 | a=36.76
c=36.76
c=35.76 | b=35
a=37.76

Output:- a: 37.76
b: 35
c: 36.76

# Relation operators:-

| operator | Use | Discreptions: |
|---|---|---|
| 1. ">" | $\boxed{op1 > op2}$ | op1 is greater than op 2 |
| 2. ">=" | $\boxed{op1 >= op2}$ | op1 is greater than or equal to op2 |
| 3. "<" | $\boxed{op1 < op2}$ | op1 is less than op 2 |
| 4. "<=" | $\boxed{op1 <= op2}$ | op1 is less than or equal op2 |
| 5. "==" | $\boxed{op1 == op2}$ | op1 and op2 are equal |
| 6. "!=" | $\boxed{op1 != op2}$ | op1 and op2 are not equal. |

The outcome of these operation is a Boolean Value.

# Conditional Statements:-

→ Any meaningful expression is called a statement.

## If Statement:-

Syntax:-

```
int main(){
    if(condition) {
        // Code to be executed
    }
    Rest of the code
}
```

Ex:1 int main(){

```
int num;
cin>> num;
if(num>0){
    Cout<<"positive";
  }
}
```

Output:-
cin= 5
positive

else-if statement Syntax

```
int main(){
    if (condition){

    }
    else if (condition){

    }
}
```

Ex:-

```
int main(){
  int a;
  Cout<<"Enter your number: ";
  cin >>a;
  if(a>0){
      Cout<<"positive"<<endl; }
  else if(a<0){
      cout<<"negative"<<endl; }
```

Output
cin = 5
positive

# How to check a number is positive or negative?

(1)

```
#include <iostream>
using namespace std;
int main(){
    int a;
    cout<<"Enter a number:";
    cin>>a;
    if(a>=0){
      cout<<a<<" is a positive number"<<endl;}
    else{ cout<<a<<" is a negative number"<<endl;}
}
```

Output:-

Enter a number: 10

10 is a positive number

Output:-

Enter a number: -5

-5 is a negative Number.

**Note:-** Condition -**এর সময়** Always ()-এর bracket **ব্যবহার করতে হবে।**

(2) int main(){

int a;
Cout<<"Enter a number:";
cin>>a;
if(a<0){
Cout<<a<<" is a negative number"<<endl;}
else{ Cout<<a<<" is a positive number"<<endl;}

}

**Output:-**

Enter a number: 100

100 is a positive number

**Output:-**

Enter a number: -50

-50 is a negative number

# How to Check a Number is a even||odd number:-

→ To check even or odd we need to look at the Conditions:-

1. Even Condition: ($x\%2 == 0$) where $x$ = input value
2. Odd Condition: ($x\%2 != 0$)

(1)
```
#include<iostream>
using namespace std;
int main(){
   int a;
  Cout<<"Enter a Number: ";
  Cin>>a;
  if(a%2==0){
  Cout<<a<<"is a even number"<<endl;}
else{ cout<<a<<" is a odd number"<<endl;}
}
```

Output:-

Enter a number: 5
5 is a odd number

Output:-

Enter a number: 10
10 is a even number.

(2) int main (){

int a;

Cout <<"Enter a number:";

cin >> a;

if(a%2 != 0){

Cout<<a<<" is a odd number"<<endl; }

else{ cout<<a<<is a even number"<<endl;}

}

Output:-

Enter a number: 7

7 is a odd number

Output:-

Enter a number: 18

18 is a even number.

# How to check a number is divisible by 'x' number

→ The conditions for checking whether a number divisible by 'x' number.

1. $(a \% x == 0)$

Check for 5:

```
#include<iostream>
using namespace std;
int main(){
 int a;
 cout<<"Enter a number:";
 cin>>a;
 if(a%5==0){
   cout<<a<<" is divisible by 5"<<endl;}
 else{ cout<<a<<" is not divisible by 5"<<endl;}
}
```

Output:-

Enter a number: 20
20 is divisible by 5

Output:-

Enter a number: 23
23 is not divisible by 5.

# How to check if the square of a number is less than 20

→ The condition is

1. (a*a <20)

```
#include <iostream>
using namespace std;
int main(){
   int a;
   cout<<"Enter a number:";
   cin>>a;
   if(a*a <20){
      cout<<a*a <<"is less then 20"<<endl;}
   else{ cout <<a*a<<"is not less then 20"<<endl;}
}
```

**Output:—**

Enter a number: 4
16 is less then 20.

**output:—**

Enter a number: 5
25 is not less then 20

→ we can use the build function instead of doing it manually:—

```
#include<iostream>
#include<math.h>
using namespace std;
int main(){
int num;
cout<<"Enter a number :";
cin>> num;
int sq= pow(num, 2);
if (sq<20){
cout<<sq<<" is less than 20"<<endl;}
else{ cout<<sq<<" is not less then 20"<<endl;}
}
```

Output:—

Enter a number: 5

25 is not less than 20

Output:—

Enter a number: 2

4 is less then 20.

# Multiple Conditions-

=> যদি কখনো আমাদের একের অধিক Condition ব্যবহার করা লাগে সে ক্ষেত্রে আমরা else if ব্যবহার করে থাকি।

## Write a c++ Code that will print hi for a hello for b, and bye for c.

```
# include <iostream>
using namespace std;
int main(){
    char var1;
    char var2;
    char var3;
    Cout<<"Enter a or b or c:";
    Cin>> var1;
    if (var1 == a){
        Cout<<"Hi"<<endl;}
    else if(var1 == b){
        Cout<<"Hello"<<endl;}
    else if (var1 == c){
        Cout<<"bye"<<endl;}
    else{ Cout<<"Invalid"<<endl;}
```

Output:-

Enter a or b or c: d
Invalid.

Output:-

Enter a or b or c: b
Hello.

1. # Write a c++ Code where given an input check if it is divisible by 5, 11 and 13

```
int main(){
    int num1;
    Cout<<"Enter your number: ";
    cin>>num1;
    if(num1%5==0){
        Cout<<num1<<"divisible by 5"<<endl; }
    else if(num1%11==0){
        Cout<<num1<<"divisible by 11"<<endl; }
    else if(num1%13==0){
        Cout<<num1<<"divisible by 13"<<endl; }
    else{ }
```

Output:-

Enter your number: 25

25 is divisible by 5 .

Output:-

Enter your number: 22

22 is divisible by 11

Notes:-

1. যদি if এবং else if value একই থাকে তাহলে if এর Condition Apply হবে।

2. আর যদি সবগুলো Condition print করতে চাই তাহলে সেক্ষেত্রে else if এর পরিবর্তে শুধু if ব্যবহার করতে হবে।

3. একসাথে অনেকগুলো if থাকলে সেক্ষেত্রে কেবল মাত্র শেষের if এর জন্য else condition কাজ করবে বাকি গুলোর জন্য else কাজ করবে না।

# Nested if/else:-

→ যখন একটি if/else if এর ভিতরে আরেকটি if/else ব্যবহার করি তখন তাকে বলা হয় Nested if/else.

# Write a c++ code to check a year is a leap year or not

→ The Condition is:-

1. year%400 == 0
2. year % 4 == 0 and year % 100 != 0

```
# include <iostream>
using namespace std;
int main(){

    int year;
    cout<<"Enter a year:";
    cin >> year;
    if(year%400 == 0){
cout<< year<<"is a leap year"<<endl;}
    else if(year%4==0){
      if(year%100 !=0){
        cout<< year<<" is a leap year"<<endl;}
      else{ cout<<year<<"is not a leap year"<<endl;}
    }

else{ cout<<year<<"is not a leap year" <<endl;
}
```

**Output:-**

Enter a year: 1980

1980 is a leap year

**Output:-**

Enter a year: 2022

2022 is not a leap year.

## # Logical operators:-

| operator | | meaning |
|---|---|---|
| 1. && | → | Logical and |
| 2. \|\| | → | Logical OR |
| 3. ! | → | Logical NOT |

| A | B | And | OR |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

&& → যদি দুইটি Condition এর মধ্যে && ব্যবহার করা হয় তাহলে যে কোন দুইটি Condition ই যদি true হয় তাহলে কেবল Statement টি Apply হবে, আর যদি একটিও ভুল থাকে তাহলে হবে না।

|| → যদি দুইটি condition এর মধ্যে || ব্যবহার করে হয় তাহলে ক্ষেত্রে condition দুইটির মধ্যে যদি যে কোন একটি true হয় তবে যে কোন statement টি Apply হবে, আর যদি দুইটিই ভুল থাকে তাহলে হবে না।

## # Complete leap year code using logical operator

```
# include<iostream>
using namespace std;
int main(){
   int year;
   cout<<"Enter a year:";
   cin>>year;
   if(year%400==0 ||(year%4==0 && year%100 != 0)){
   cout<<year<<" is a leap year"<<endl;}
else{cout<<year<<" is not a leap year"<<endl;}

}
```

Output:-
Enter a year: 2023
2023 is not a leap year

# Write a c++ code where the inputs are num1 and num2 and are operators @, #, $, &
where operator @ means = sum
" # " = sub
" $ " = div
" & " = mult

=>
```
#include<iostream>
using namespace std;
int main(){
  float num1, num2;    char op;
  cout<<"Enter your numbers:";
  cin>>num1>>num2;
  cout<<"Enter your operator:";
  cin>>op;
  if(op='@'){
      result = num1+num2;
  cout<< "Result:"<<result<<endl; }
  else if(op='#'){
      result= num1-num2;
  cout<<"Result:"<<result<<endl;}
  else if(op='$'){
      result=num1/num2;
  cout<<"Result:"<<result<<endl;}
```

```
else if(op = '*'){
    result = num1 * num2;
  Cout<<"Result:"<<result<<endl; }
else { Cout<< "Invalid operator"<<endl; }
}
```

Output:

Enter your numbers: 5, 2
Enter your operator: *
Result: 10

# #Switch case:

→ যদি অনেকটা if/else এর মত কাজ করে, তবে মানে হচ্ছে আমরা অনেক গুলো if/else এর পরিবর্তে Switch Case ব্যবহার করতে পারি

1: Switch statement provides a better Alternative than a large series of if-else statement

2: Switch statement executes one statement form multiple condition.

3: Keywords used in switch statement:
Switch, case, break, default.

Syntax-

```
switch (expression) {
Case value 1:
    //Codes to be executed
    break;
Case value 2:
    //Codes to be executed
    break;
Case ValueN:
    //Codes to be executed
    break;
default:
    // Codes to be executed
```

# Complete previous code using Switch Case Statement:-

```
#include <iostream>
using namespace std;
int main() {
int    num1;
int    num2;
    char OP;
```

p.T.O

```
Cout << "Enter your 2 numbers: ";
Cin >> num1 >> num2;
Cout << "Enter your operators: ";
cin >> op;
Switch (op) {
 Case '@':
   result = num1 + num2;
   Cout << "Result: " << result << endl;
   break;
 Case '#':
   result = num1 - num2;
   Cout << " Result: " << result << endl;
   break;
 Case '$':
   resultt = num1 / num2;
   Cout << "Result: " << result << endl;
   break;
 Case '&':
   result = num1 * num2;
   Cout << " Result: " << result << endl;
   break;
 default:
   Cout << "In Valid operator" << endl;
}
```

Output:-

Enter your 2 numbers: 10, 2
Enter your operator: @
Result: 102

Output:-

Enter your 2 numbers: 11, 5
Enter u operator: #
Result: 6

**Note:-** Switch এর মধ্যে float ব্যবহার করা যায় না শুধুমাত্র int ব্যবহার করা হয়।

## #Loop:-

⇒ A loop in c++ is a programming construct that allows me to execute a block of code repeatedly until a certain condition is met.

There are three types of loop:-

1. for loop
2. while loop
3. do-while loop.

# For loop:

Syntax:-

initialization ↗ Condition ↗ Increment/decrement

```
for( int i=1; i=5; i++) {
    Cout<<"Bangladesh" <<endl;
```

So, Syntax is:-

```
for(initialization, Condition, increment/decrement) {
    Statement(s)
}
```

# How to print at a time 5 Numbers

```
#include <iostream>
using namespace std;
int main () {
    for(int i=0; i<5; i++) {
        int number;
        Cout << " Enter a number: ";
        cin >> number;
    }
}
```

Output:-

Enter a number:1
Enter a number:2
Enter a number: 3
Enter a number:4
Enter a number:5

# Check 5 numbers at a time is a even number || odd number:-

```
# include<iostream>
using namespace std;
int main(){
for ( int i=1; i<=5; i++){
  int num;
  Cout <<"Enter a number :";
  Cin>>num;
  if(num%2==0){
    Cout<<Num <<"is a even number"<<endl;}
else{ Cout<< Num<<"is a odd number"<<endl;}
}
}
```

Output:
Enter a number:5
5 is a odd number

P-T-O

Enter a number : 4
4 is a even number
Enter a number : 40
40 is a even number
Enter a number : 7
7 is a odd number
Enter a number : -52
-52 is a even number.

# How to check 5 numbers at a time is a positive number or negative

```
#include <iostream>
using namespace std;
int main(){
for(int i=1; i<=5; i++){
    int num;
    Cout<<"Enter a number :";
    cin >> num;
    if(num>=0){
        Cout<<num<<"is a positive number"<< endl; }
    else{ Cout<<num<<" is a Negative number"<<endl; }
}
}
```

Output:-

Enter a number: 1
1 is a positive number
Enter a number: -1
-1 is a negative number.
Enter a number: 0
0 is a positive number
Enter a number: 400
400 is a positive number
Enter a number: -4
-4 is a negative number.

# Do Sum in for loop:-

```
int main(){
    int num;
    int sum=0;
    for(int i=1 ; i<=5 ; i++){
        cout<<"Enter a number:";
        cin >> num;
        sum= sum+ num; }
    cout<<"sum: "<<sum<<endl;
}
```

এইক্ষেত্রে: এক সাথে সবগুলো sum দেখতে চাই, cout<<"sum:" for loop এর বাইরে বসানো হয়েছে.

output:-

Enter a number: 5
Enter a number: 4
Enter a number: 3
Enter a number: 2
Enter a number: 1

Sum: 15

## #Loop এর Actual ব্যবহার:-

### #Check any number is prime number or not using loop:-

```
#include<iostream>
using namespace std;
int main() {
    int num;
    Cout<<"Enter a number: ";
    Cin >> num;
    int prime = 0;
    for (int i=2; i<num; i++) {
        if (num%i ==0) {
            prime = 1;
            break; }
        else { prime=0 }
    }
    if (prime==1) {
        Cout<< num <<" is not a prime number" <<endl;
```

else{ cout<<num<<"is a prime number";}

Output:-

Enter a number: 3

3 is a prime number

Dry run:

25 i=2

if(25%2==0) ×

prime=0

i=3

if(25%3==0) ×

prime=0

i=4

if(25%4==0) ×

prime=0

i=5

if(25%5==0) ✓

prime=1

Loop break;

not prime.

5 i=2

if(5%2==0) ×

prime=0

i=3 (5%3==0) ×

prime=0

i=4 (4%3==0) ×

prime=0

is prime number.

# While loop:-

## Syntax:-

```
initialization;
while(condition){
    Statement(s)
    increment/Decrement
}
```

## Print 0 - 9 using while loop:-

```
#include<iostream>
using namespace std;
int main(){
int i=0;
while(i<5){
    cout<<"i:"<<i<<" ";
    i++;
}
}
```

**Output:**

i: 0 i: 1 i: 2 i: 3 i: 4

# print 1-5 using same condition in while loop:-

```
#include <iostream>
using namespace std;
int main(){
    int i=0;
    while(i<5){
        i++;
        cout<<"i:"<<i<<" "; }
}
```

**Output:-**

i:1 i:2 i:3 i:4 i:5

# What happens if the (<) is sign is reversed:-

```
int main(){
    int i=0;
    while(i>5){
        cout<<"i:"<<i<<" ";
        i++; }
}
```

**Output:-**

Nothing / cz. Condition ভুল আসবে

# do-while loop:-

Syntax:-

```
initialization;
do { Statement(s)
     increment/decrement;
   }
while(condition);
```

# print 0-4 using do-while loop:-

```
=> #include <iostream>
   using namespace std;
   int main() {
     int i=0
     do { cout<<"i:"<<i<<" ";
          i++ }
     while (i<5);
   }
```

Output:-

i:0 i:1 i:2 i:3 i:4

Q: What is the difference between while and do-while loop?

⇒ The main difference between a while loop and do-while loop.

1. Loop is that in a while loop, the condition is checked at the beginning of the loop, and if it is false, the loop will never execute, wheres a do-while loop will always execute at least once before checking the condition.

# Convert for to while loop previous prime number check code in c++

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    int prime = 0;
    int i = 2;
    while (i<num) {
```

P.T.O

```
if(num % i == 0){
    prime = 1;
    break; }
else { prime = 0; }
i++;
}
if (prime == 0){
   Cout<<num<<"is a prime number"<<endl; }
else{ Cout<<num<<"is not a prime number"<<endl; }
}
```

Output:-

Enter a number: 8

8 is not a prime number.

Output:-

Enter a number: 7

7 is a prime number.

Convert for loop to do-while loop

previous prime number check code:

```
# include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a Number: ";
    cin >> num;
    int prime = 0;
    int i = 2
    do { if (num % i == 0) {
            prime = 1;
            break; }
         else { prime = 0; }
    }
    i++
    while (i < num);
    if (prime = 0) {
        cout << num << " is a prime number" << endl; }
    else { cout << num << " is not a prime number" << endl; }
```

output:-

Enter a number: 21

21 is not a prime number.

# Swap Operation

```
# include<iostream>
using namespace std;
int main(){
  int a=5;
  int b=10;
    Cout<<"a:"<<a <<endl;

    Cout<<"b:"<< b << endl;
  int c= a;
  Cout<<"After swap operation"<<endl;
  a=b;
  b=c;
  Cout<<"a :"<<a <<endl;
  Cout<<"b:"<<b<<endl;
```

***

**Q:** What is break in C++?

=> "break" in c++ is a keyword used to exit a loop premantarely, regardless of whether of the loop's Condition has been fully Satisfied.

**Q:- What is Continue in C++?**

⇒ In C++, "Continue" is a key word used to skip the remaining statements in the current iteration of a loop and move on to the next iteration.

**Q:- What is the difference between Continue and break:**

⇒ In C++, "Continue" is used to skip the remaining statements in the current iteration of a loop and move on the next iteration,·
While "break" is used to exit the loop · paremataraly, regardless of whether the loop's condition has been fully satisfied.

## # Continue:-

```
#include <iostream>
using namespace std;
int main() {
    for (int i=0; i<5; i++) {
        if (i==3) {
            Continue; }
    Cout <<i<< " "; }
}
```

**Output:-**

0 1 2 4

```
# include <iostream>
  using namespace std;
  int main(){
    for(i=0;  i<5;  i++){
    Cout<<i<<" ";
    if(i==3){
      Continue;}
    }
  }
```

Output:-

0 1 2 3 4

# Nested for loops:-

⇒ for-loop এর মধ্যে আরেকটি for loop থাকলে তাকে Nested for loops বলা হয়।

⇒

```
# include <iostream>
  using namespace std;
  int main(){
   for (int i=0;  i<3;  i++){
     for (int j=0;  j<2;  j++){
      Cout<< "Hello";}
     Cout<<endl;}
  }.
```

Output:-

Hello Hello
Hello Hello
Hello Hello

ব্যাখ্যা:-

i=0
j=0 → Hello Hello
j=1

i=1
j=0 → Hello Hello
j=1

i=2
j=0 Hello Hello
j=1

# How to print a half pyramid:-

```
#include <iostream>
using namespace std;
int main(){
    for(int i=0; i<5; i++){
        for(int j=0; j<=i; j++){
            Cout<<"*"; }
    cout<<endl; }
}
```

Output:-

```
*
**
***
****
*****
```

ব্যাখ্যা:-

| | | |
|---|---|---|
| i=0 | j=0 | * |
| i=1 | j=0,1 | ** |
| i=2 | j=0,1,2 | *** |
| i=3 | j=0,1,2,3 | **** |
| i=4 | j=0,1,2,3,4 | ***** |

# What if user wants to input rows:-

```
#include<iostream>
using namespace std;
int main(){
    int row;
    cout<<"Enter your rows number:" ;
    cin>>row;
    for(int i=0; i<row; i++){
    for(int j=0; j<=i; j++){
        cout<<"*" ;}
    cout<<endl;}
}
```

Output:-

Enter your rows: 7

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
```

SIRO

i=0    j=0    *
i=1    j=0,1    * *
i=2    j=0,1,2    * * *
i=3    j=0,1,2,3    * * * *

# How do we print the inverted pyramid in c++?—

```
#include <iostream>
using namespace std;
int main(){
    int row;
    cout<<"Enter the row :";
    cin>>row;
    for( int i=0 ; i<row ; i++){
        for( int j=row; j>i; j--){
            cout<<"*"; }
        cout<<endl; }
}
```

Output—

Enter the row: 5
* * * * *
* * * *
* * *
* *
*

Output—

i=0 j=1,2,3,4,5 → * * * * *
i=1 j=1,2,3,4 → * * * *
i=2 j=1 2 3 → * * *
i=3 j=1 2 → * *
i=4 j=1 → *

# How to print full pyramid in c++

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 0; i<5; i++) {
        for (int j = 5; j>i; j--) {
            cout<<" ";
        }
        for (int k=0; k<(2*i+1); k++) {
            cout<<"*";
        }
        cout<<endl;
    }
}
```

**Output:-**

```
    *
   * * *
  * * * * *
* * * * * * * *
```

star

$i=0$
$2i+1 = 2*0+1 = 1$

$i=1$
$2i+2 = 2*1+1 = 3$

$i=2$
$2i+2 = 2*2+1 = 5$

$i=3$
$2i+2 + 2*3+1 = 7$

$i=4$
$2i+2 + 2*4+1 = 9$

# How to print inverted full pyramid in C++ :-

```
#include <iostream>
using namespace std;
int main() {
    int row;
    Cout << "Enter rows: ";
    Cin >> row;
for( int i=0;  i<row;  i++){

    for( int j=0;  j<=i;  j++){
        Cout<<" "; }
for( int k= row;  k>=(2*i+1);  k--){
    Cout<<"*"; }
Cout <<endl; }
```

Output:-

Enter rows: 5

```
* * * * *
  * * *
    *
```

Agains

```
#include<iostream>
using namespace std;
int main(){
  int n;
  Cout<<"Enter the Height";
  cin>>n;
 for (int i=n;  i>=1  i--){
  for(int j=1; j<=n-i; j++){
     Cout<<" ";}
 for (int k=1;  k<=(2*i-1) ; k++){
   Cout<<"*";}
 Cout<<endl;
 }
}
```

Output:-

Enter the Height :- 5

```
*********
 *******
  *****
   ***
    *
```

i=5

$k=(2\times5-1)=9 \mid j=5-5=0$

i=4

$k=(2\times4-1)=7 \mid j=5-4=1$

i=3

$k=(2\times3-1)=5 \mid j=5-3=2$

$k=(2\times2-1)=3 \mid j=5-2=3$

i=1

$k=(2\times1-1)=1 \mid j=5-1=4$

# Array :-

=> এক সাথে অনেকগুলো Value store করার জন্য Array ব্যবহার করা হয়।

Syntax:-

name of the array

int arr[50]

datatype

Size of the array

# How are the Values stored in Arrays :

```
int main(){
    int arr[5] = {10, 20, 65, 70, 5};
```

index :-

| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 10 | 20 | 65 | 70 | 5 |

so, arr[3] = 70

arr[0] = 10

arr[4] = 5

arr[1] = 20

```
#include <iostream>
using namespace std
int main(){
    int arr[5] = {10, 20, 65, 70, 5};
    cout << arr[4] << endl;
    cout << arr[2] << endl;
}
```

**Output:-**

5
65

## # How to measure the size of Arrays:-

```
#include <iostream>
using namespace std;
int main(){

    int arr[5] = {10, 20, 30, 40, 50};
    cout << size of (arr);
}
```

**Output:-**

20

size
int = 4
arr = 4×5 = 20

# How to print All Array at a time using loop:—

```
#include <iostream>
using namespace std;
int main(){
    int arr[5] = {2,3,5,6,7};
    Cout<<"Elements in Array:" ;
    for (i=0; i<5; i++){
        Cout<<arr[i] <<" "; }
}
```

**Output:—**

2, 3 5 6 7

# If you want to take input from the user:—

```
int main(){
    int arr[6];
    Cout<<"Enter the Elements:";

    for( i=0; i<6; i++){
        cin >> arr[i]; }
```

```
cout<<"Elements in Array:";
 for( int i=0 ;  i<6;  i++){
   cout<<arr[i]<<" ";}
}
```

Output:-

Enter the Elements: 1 3 2 5 8
Elements in Array: 1 3 2 5 8

# How to print character and special character in Arrays:-

```
#include<iostream>
using namespace std;
int main(){
  char arr[6];
  cout<<"Enter the Elements:";
  for (int i=0; i<6; i++){
     cin>>arr[i];}
  cout<<"Elements in array:";
  for ( int i=0  i<6; i++){
  cout<<arr[i]<<" ";}
}
```

Output:-

Enter the elements: v @ # 1 2
Elements in array: v @ # 1 2

# for Double quotation:-

⇒ Double কোটেশান এর মধ্যে আমরা যেগুলোর শেষে একটি null character থাকে এই জন্য আমাদের Array এর সাইজ যা থাকবে তার থেকে 1 বেশি দিতে হবে। তা না হলে Compiler কোন error দেখাবে।

Ex:- int arr[7] = "ABC1234" → Error
int arr[8] = "ABC1234"

যখন arr[7] দিলাম তখন আমরা 7টি value input দিয়েছি যেহেতু null character এর জায়গা না থাকায় তাই Error আসবে।

- বাড়তি জায়গা বাড়িয়ে arr[8] করায় তাহলে আর কোনো Error আসে নাই।

# #How to check four numbers even or odd using array:—

```
# include <iostream>
using namespace std;
int main(){
  int arr[4];
  Cout<<"Enter the Elements: ";
  for( int i=0; i<4; i++){
       cin>>arr[i];}
  for ( int i=0; i<4 ; i++){
   if (arr [i]    %2 ==0){
     Cout<<arr[i]<<"its a even number:"<<endl;}
  else { Cout<<arr[i]<<"its a odd number"<<endl; }
  }
}
```

Output:—

Enter the elements: 1, 20, 3, 40
1 is a odd number
20 is a even number
3 is a odd number
40 is a even number

# Do Sum Using Array & Average & Check Average is even/odd

```
#include <iostream>
using namespace std;
int main(){
float arr[5] = {10.6, 5, 4.2, 6, 8};
float Sum=0;
float Avg=0;
for(int i=0; i<5; i++){
    Sum = sum+ arr[i];}
Cout <<"Sum:" << sum <<endl;
    avg = Sum/5;
Cout <<"Average:" <<avg<< endl;
if((int) avg%2==0){
  Cout<<(int) avg << "is a even number"<<endl;}
else { cout <<(int)avg << "is a odd number"<<endl;}
}
```

Output:

Sum: 33.7
Average: 6.74
6 is a even number

# How to check years is a leap year or not using array

```
# include <iostream>
using namespace std;
int main(){
  int n
  cout<<"Enter the size of array :";
  cin>> n;
  int arr[n];
  Cout<<"Enter years:";
  for (int i=0; i<n; i++){
      cin >> arr[i]; }
  for (int i=0; i<n; i++){
   if (arr[i]%400==0 || (arr[i]%4==0 && arr[i]%100!=0)){
      Cout<<arr[i]<<"is a leap year"<<endl; }
  else{ cout<<arr[i] <<"is not a leap year"<<endl;}
}
```

Output:—

Enter the size of array: 5
Enter years: 2000, 1998, 2004, 2002, 2001
2000 is a leap year
1998 is not a leap year
2004 is a leap year
2002 is a not a leap year.

2001 is not a leap year

# Write a c++ Code takes n numbers as inputs and strore them in a single array. Now find the even numbers and odd numbers from the array and store them in a separate array:—

```
#include <iostream>
using namespace std;
int main(){
    int n;
    Cout<<"Enter the size of array:";
    cin>>n;
    int arr[n];
    Cout<<"Enter your number:";
for (int i=0; i<n; i++){
    cin>>arr[i];}
int even[n];
int odd [n];
int j=0;
int k=0;
for (int i=0; i<n; i++){
    if (arr[i]%2==0){
        even[j]= arr[i];
        j++; }
```

```
else{
        odd[k] = arr[i];
        k++; }
}
Cout << "The even element: "
for (int i=0; i<j; i++){
    Cout << even[i] << " "; }
Cout << "The odd element: "
for (int i=0; i<k; i++){
    Cout << odd[i] << " "; }
}
```

Output:-

Enter the size of array: 5
Enter your numbers: 2, 3, 4, 5, 6
The even element:- 2 4 6
The odd element: 3 5